COURSE 4 · CHEMISTRY

Chemistry

Visual companion and Symbols Sheets

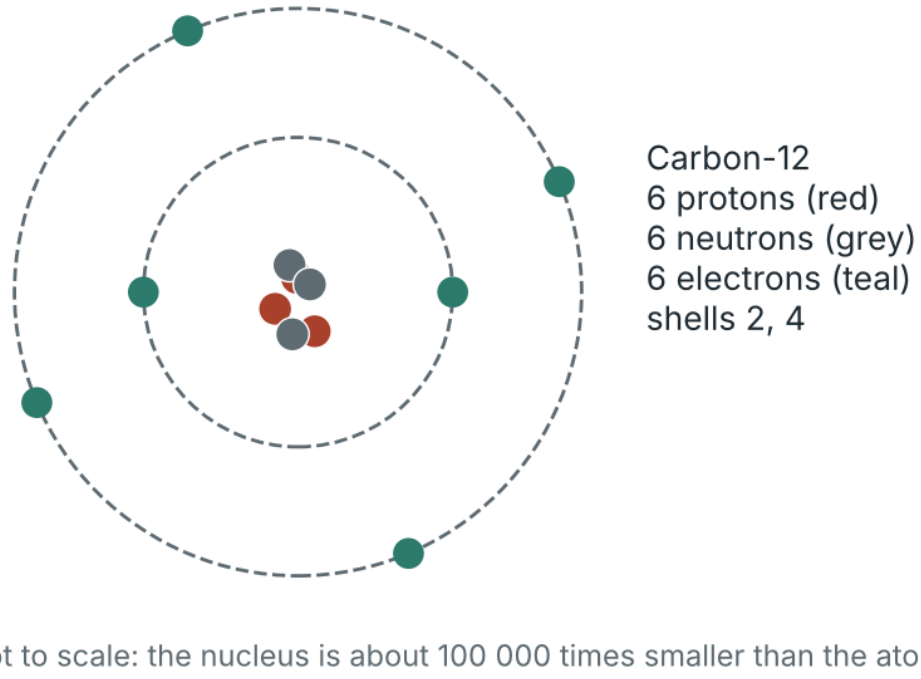
One section per module: the Symbols Sheet, with every numbered equation from the audio reader in ordinary notation, then the module's key diagram.

Keep it open while you listen, and beside you for the workbook.

Module 4.1 · Symbols and diagrams

Matter and atomic structure

4.1-1	Mass number	$A = Z + N$
4.1-2	Relative atomic mass	$A_r = \sum(\text{isotope mass} \times \text{fractional abundance})$
4.1-3	Sodium's configuration	$\text{Na}: \ 1s^2\,2s^2\,2p^6\,3s^1 \quad (2, 8, 1)$



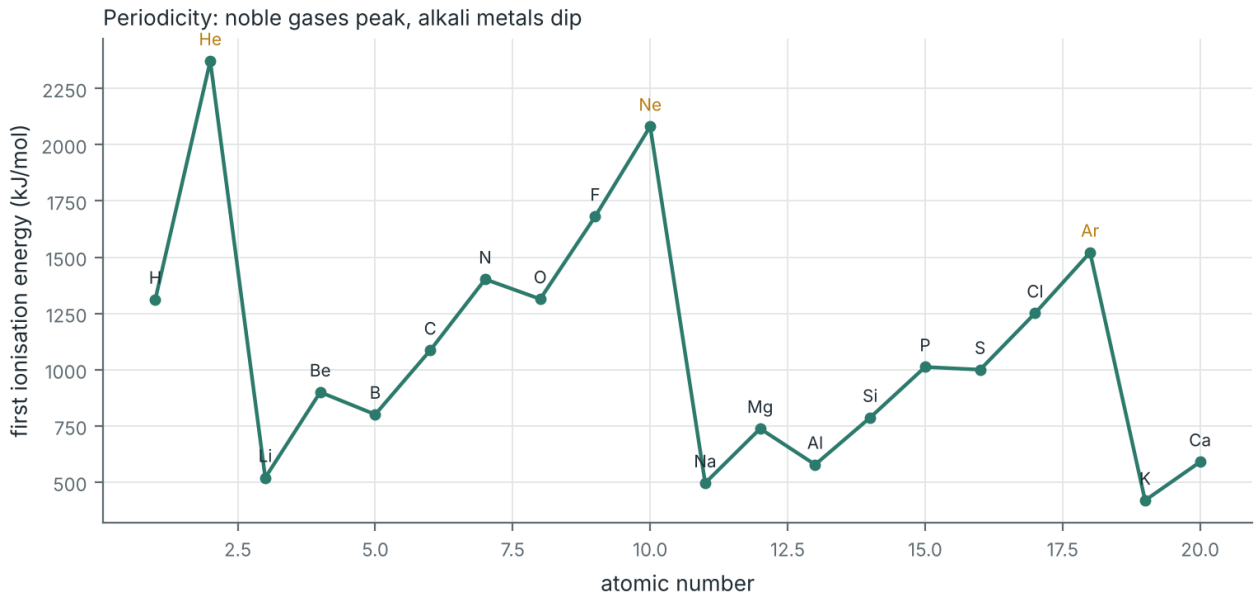
A carbon-12 atom in the shell model.

Particle	Charge	Relative mass	Where
proton	+1	1	nucleus
neutron	0	1	nucleus
electron	−1	1/1836	shells around the nucleus

Module 4.2 · Symbols and diagrams

The periodic table

4.2-1	Ionisation (first)	$\mathrm{X(g) \rightarrow X^{+}(g) + e^{-}}$
4.2-2	Alkali metal with water	$\mathrm{2Na + 2H_2O \rightarrow 2NaOH + H_2}$



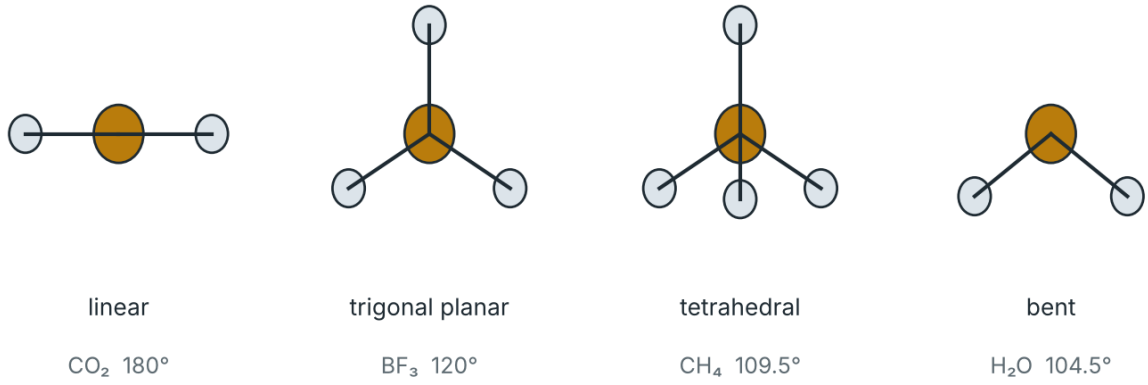
First ionisation energies of the first 20 elements. The pattern repeats each period: this is the periodicity that names the table.

Module 4.3 · Symbols and diagrams

Bonding and molecular shape

4.3-1	Ionic formation	$\mathrm{Na} + \mathrm{Cl} \rightarrow \mathrm{Na}^{+} + \mathrm{Cl}^{-}$
4.3-2	Covalent sharing in water	$\mathrm{H{-}O{-}H}$

Electron pairs repel and spread out as far as possible (VSEPR)



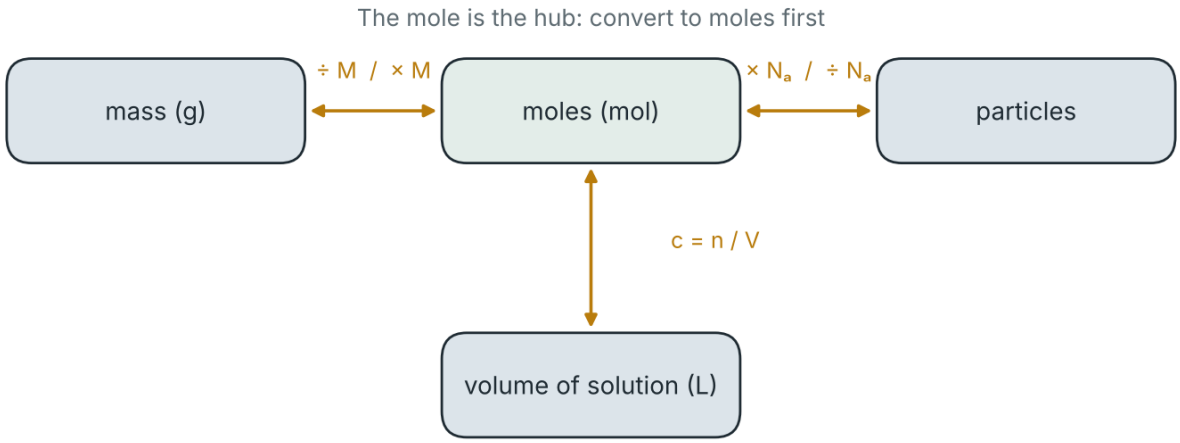
linear	trigonal planar	tetrahedral	bent
CO_2 180°	BF_3 120°	CH_4 109.5°	H_2O 104.5°

Common molecular shapes predicted by VSEPR.

Module 4.4 · Symbols and diagrams

The mole and stoichiometry

4.4-1	Moles from mass	$n = \frac{m}{M}$
4.4-2	Number of particles	$N = nN_A, \quad N_A = 6.02 \times 10^{23}\ \text{mol}^{-1}$
4.4-3	Combustion of methane	$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
4.4-4	Percentage yield	$\%\ \text{yield} = \frac{\text{actual}}{\text{theoretical}} \times 100$
4.4-5	Concentration and dilution	$c = \frac{n}{V}, \qquad c_1V_1 = c_2V_2$

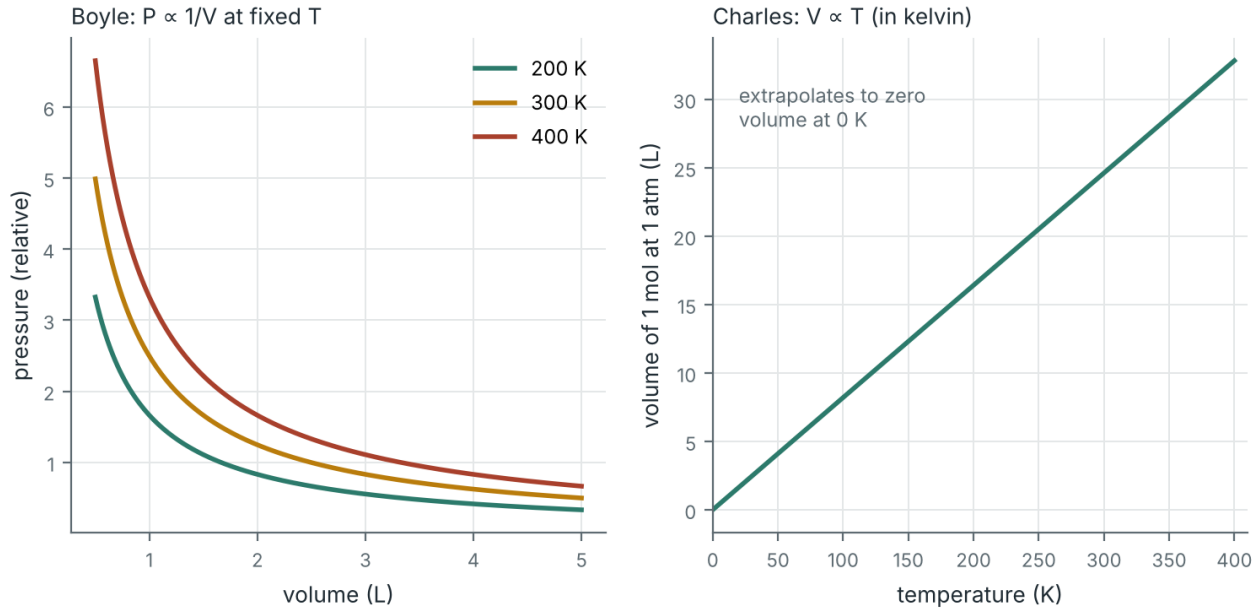


The mole as the central conversion hub.

Module 4.5 · Symbols and diagrams

States of matter and gas laws

4.5-1	Boyle's law	$P_1V_1 = P_2V_2$
4.5-2	Charles' law	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
4.5-3	Ideal gas law	$PV = nRT, \quad R = 8.314\ \mathrm{J\,mol^{-1}K^{-1}}$
4.5-4	Mean kinetic energy	$\overline{E_k} = \frac{3}{2}k_BT$

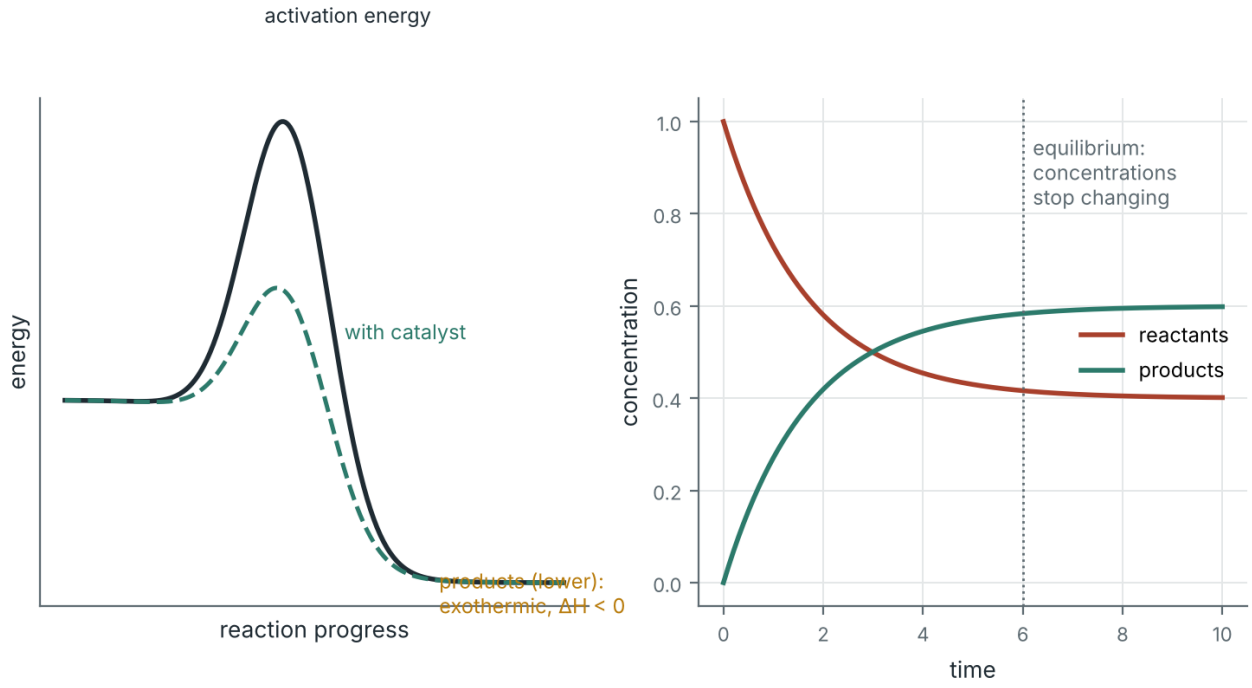


Left: Boyle's law curves at three temperatures. Right: Charles' law, volume proportional to kelvin temperature.

Module 4.6 · Symbols and diagrams

Energy, rates and equilibrium

4.6-1	Enthalpy change	$\Delta H = \sum E_{\text{bonds broken}} - \sum E_{\text{bonds made}}$
4.6-2	Equilibrium constant for aA + bB ⇌ cC + dD	$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$
4.6-3	Haber process	$\mathrm{N_2} + 3\mathrm{H_2} \rightleftharpoons 2\mathrm{NH_3}, \quad \Delta H < 0$

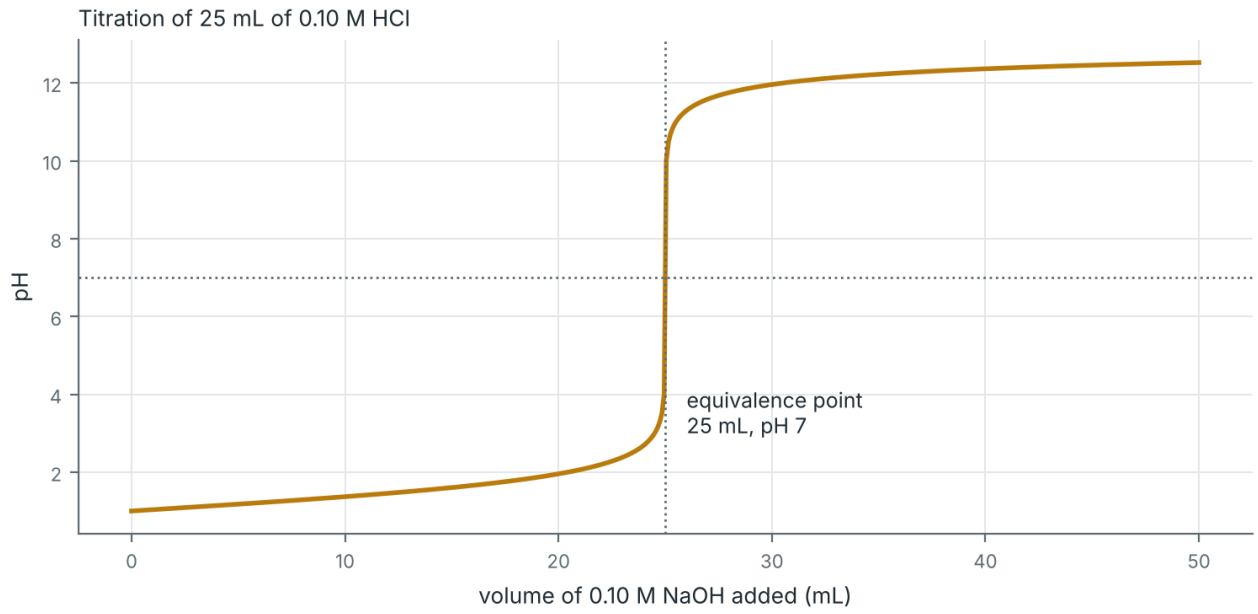


Left: an energy profile with and without a catalyst. Right: approach to equilibrium.

Module 4.7 · Symbols and diagrams

Acids, bases and redox

4.7-1	pH	$\mathrm{pH} = -\log_{10}[\mathrm{H}^+]$
4.7-2	Neutralisation	$\mathrm{HCl} + \mathrm{NaOH} \rightarrow \mathrm{NaCl} + \mathrm{H_2O}$
4.7-3	A redox pair	$\mathrm{Zn} + \mathrm{Cu}^{2+} \rightarrow \mathrm{Zn}^{2+} + \mathrm{Cu}$
4.7-4	Water's ion product	$[\mathrm{H}^+][\mathrm{OH}^-] = 10^{-14}$ at $25\ ^\circ\mathrm{C}$



A titration curve: pH jumps sharply at the equivalence point.

Module 4.8 · Symbols and diagrams

Organic chemistry essentials

4.8-1	Alkane formula	$\mathrm{C_nH_{2n+2}}$
4.8-2	Esterification	$\mathrm{acid + alcohol \rightleftharpoons ester + water}$
4.8-3	Addition polymerisation	$n\,\mathrm{CH_2{=}CH_2 \rightarrow (CH_2{-}CH_2)_n}$

alkane	alcohol	carboxylic acid	amine	ester
C–C, C–H only	–OH	–COOH	$–NH_2$	–COO–
ethane	ethanol	ethanoic acid (vinegar)	methylamine	fruit flavours, fats

A functional group decides how a molecule reacts; the carbon skeleton is the scaffold it hangs on.

Common functional groups and everyday examples.